

Vishakha Agrawal

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EDUCATION

International Institute of Information Technology, Hyderabad (IIIT-H)

Bachelor of Technology in Computer Science (Third Year)

Hyderabad, India

July 2023 – Present

COURSEWORK

Computer Science: Computer Programming, Introduction to Software Systems, Data Structures and Algorithms, Computer Systems Organization, Operating Systems and Networks, Algorithm Analysis & Design, Automata Theory, Design and Analysis of Software Systems, Introduction to Information Security

Data Science & ML: Data and Applications, Machine Data and Learning, Machine Learning for Natural Sciences

Geospatial & Remote Sensing: Introduction to Spatial Sciences, Spatial Informatics, Advanced Remote Sensing, Optical Remote Sensing, Spatial Data Sciences

Behavioral & Cognitive Sciences: Introduction to Brain and Cognition, Introduction to Neuroeconomics, Behavioral Research: Experimental Design

Digital Systems & IoT: Digital Systems and Microcontrollers, Introduction to IoT, Embedded Systems Workshop

Mathematics: Discrete Structures, Linear Algebra, Real Analysis, Probability and Statistics

SELECTED PROJECTS

Microbiome-Sparing Antibiotic Discovery Pipeline | *PyTorch, Chempromp, RDKit* [GitHub](#)

- Five-model ML pipeline (RF, D-MPNN, CheMeleon, MoLFormer-XL, D-MPNN+RDKit) to identify antibiotics that kill pathogens while sparing gut commensals. Trained on 108,625 instances across 7 tasks with scaffold CV.
- Screened 6,739 Drug Repurposing Hub compounds. Consensus: 890 candidates, 3 with 5/5 model agreement. Validated against Stokes et al. (Cell, 2020).

IoT-Enabled Smart Refill Pod Monitoring | *Flutter, Dart, Firebase* Work for Live Customer (Proprietary)

- Real-time mobile dashboard for monitoring IoT refill pods: live device status, inventory, temperature/humidity, and automated offline detection with configurable per-kiosk alerting.
- Role-based access control, remote command dispatch to kiosks, and environmental threshold alerts.

Buy-Sell E-Commerce Platform | *MERN Stack, JWT* [GitHub](#)

- Full-stack MERN app for IIITH buy/sell with authentication, shopping cart, search/filtering, and JWT sessions.

Distributed Network File System | *C, Sockets, TCP/IP* [GitHub](#)

- Client-server distributed file system with naming server, trie-based path resolution, mutex locks, LRU caching, and async writes.

Enhanced XV6 Operating System | *C, POSIX Threads, Socket Programming* [GitHub](#) [GitHub](#)

- COW fork, custom schedulers (LBS, MLFQ), TCP/UDP networking, and multi-threaded file manager with distributed sorting.

Custom Unix-like Shell | *C* [GitHub](#)

- Unix shell with pipes, I/O redirection, background/foreground process management, signal handling, and man page fetching.

Photo-to-Video Web App | *Flask, MoviePy, OpenCV, MySQL, JWT* [GitHub](#)

- Web app that converts uploaded photos into a video with music. Flask backend, JWT auth, media storage with preview.

Additional Projects

- Predictive Coral Reef Vulnerability Index (pCRVI) | Python, GEE, 7-component satellite early warning system [GitHub](#)
- Hybrid CA-MLR-GA Urban Growth Simulation | Python, GEE, Landsat [GitHub](#)
- Urban Heat Island Effect Analysis | Python, GEE, LST remote sensing [GitHub](#)
- Social Proof and AI Content Detection | Next.js, React, Python, 2×2 experiment (N=35) [GitHub](#)
- Job Scam Victimization Analysis | Python, HuggingFace Transformers, survey design (N=96) [GitHub](#)
- Word Superiority Effect Replication | PsychoJS, Pavlovia (N=41) [GitHub](#)
- Veterinary Clinic Management System | MySQL, ER design, stored procedures [GitHub](#)

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL, Dart, Bash, HTML, CSS

ML/DL: PyTorch, scikit-learn, HuggingFace Transformers, Chemprop, RDKit, pandas, NumPy, scipy, seaborn, matplotlib

Web & Mobile: React.js, Next.js, Node.js, Express.js, Flask, Flutter, Firebase, MongoDB, MySQL, PostGIS, REST APIs

Geospatial: Google Earth Engine, QGIS, PostGIS

Experiment Platforms: PsychoJS/Pavlovia